AidFlow AI

Unified Autonomous Infrastructure for Transparent Donations & Disaster Relief

> **Master Project Documentation**

Document Information

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Document Purpose

This document serves as the complete technical and functional documentation for **AidFlow AI**, covering the project's motivation, research background, architecture, workflows, features, business logic, user roles, artificial intelligence, blockchain integration, audit mechanisms, security, scalability, and future roadmap.

Unlike installation or deployment guides, this document focuses on **why the project exists, how it works internally, and how every component interacts within the system.**

It is intended for:

- Professors
- Judges
- Recruiters
- Researchers
- Developers
- Contributors
- Investors
- Technical Reviewers

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1. Introduction

Modern humanitarian aid systems process billions of dollars every year through governments, NGOs, charities, CSR organizations, foundations, and disaster relief agencies. Despite these resources, millions of beneficiaries continue to experience delayed assistance, financial leakages, corruption, inefficient verification, and a complete lack of transparency regarding how donated funds are actually utilized.

The issue is not the absence of goodwill or funding. Instead, the root cause lies in fragmented execution systems that rely heavily on manual approvals, trust-based financial transfers, isolated databases, and retrospective audits.

AidFlow AI was conceived to fundamentally redesign this execution model.

Rather than building another fundraising application, AidFlow AI introduces an integrated humanitarian execution infrastructure where policies define intent, Artificial Intelligence automates decision-making, blockchain ensures immutable accountability, and programmable financial controls enforce purpose-bound aid distribution.

The objective is to transform humanitarian trust from subjective reputation into measurable and verifiable evidence.

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2. Executive Summary

AidFlow AI is a research-driven humanitarian infrastructure designed to modernize donation management and disaster relief through intelligent automation, transparent financial execution, and immutable auditing.

The platform integrates Artificial Intelligence, Blockchain, Rule-Based Automation, Cryptographic Verification, Queue-Based Processing, and Dynamic Trust Scoring into a unified execution ecosystem.

Instead of relying solely on organizations to report how donations are spent, AidFlow AI continuously records, verifies, and audits every significant event across the donation lifecycle.

Each participant within the ecosystem—including donors, NGOs, beneficiaries, merchants, governments, and administrators—interacts through predefined workflows governed by transparent policies and supported by AI-driven validation.

The result is a platform where humanitarian accountability becomes an intrinsic property of the system rather than an optional organizational practice.

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3. Abstract

Humanitarian aid remains one of the most important global mechanisms for supporting communities affected by poverty, disasters, conflicts, and emergencies. However, existing aid systems continue to struggle with inefficient execution, fragmented governance, delayed approvals, financial misuse, and declining public trust.

AidFlow AI proposes a unified infrastructure capable of transforming the complete humanitarian execution pipeline into a programmable, transparent, and autonomous ecosystem.

The system combines:

- Artificial Intelligence
- Blockchain Technology
- Programmable Relief Wallets
- Dynamic Trust Scoring
- Rule-Based Policy Enforcement
- Immutable Audit Trails
- Multi-Role Collaboration

to create a platform where every humanitarian transaction becomes traceable, verifiable, and accountable.

Rather than functioning as a traditional donation application, AidFlow AI acts as a foundational infrastructure upon which governments, NGOs, disaster response organizations, CSR initiatives, and humanitarian agencies can build trustworthy digital aid ecosystems.

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4. Vision

To establish a globally trusted humanitarian infrastructure where every donation, relief transaction, and aid distribution is transparent, programmable, verifiable, and accountable through intelligent technologies.

AidFlow AI envisions a future in which humanitarian trust is no longer dependent on organizational reputation but is instead established through mathematical proof, immutable records, and evidence-driven governance.

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5. Mission

Our mission is to eliminate opacity, delay, and misuse within humanitarian ecosystems by integrating Artificial Intelligence, Blockchain, and programmable financial controls into one unified platform that ensures efficient, fair, and transparent aid distribution for every stakeholder.

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6. Project Overview

AidFlow AI is a full-stack humanitarian technology platform that combines multiple modern technologies into a single integrated ecosystem.

The platform consists of six major pillars:

- AI Decision Engine
- Donation Management System
- Disaster Relief Management
- Blockchain Audit Infrastructure
- Trust & Reputation Engine
- Rule-Based Financial Execution

Unlike conventional donation platforms that terminate after collecting funds, AidFlow AI manages the complete post-donation lifecycle, ensuring continuous visibility from donor contribution to beneficiary impact.

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7. Global Humanitarian Challenges

Although humanitarian organizations receive billions in annual funding, several systemic issues continue to reduce operational efficiency.

7.1 Lack of Transparency

Donors rarely receive verifiable proof regarding how their contributions are utilized.

Most organizations provide annual reports or generalized campaign updates instead of transaction-level accountability.

7.2 Delayed Aid Distribution

Traditional approval chains require multiple administrative layers before assistance reaches beneficiaries.

During disasters, even minor delays may significantly impact survival and recovery.

7.3 Financial Misuse

Cash-based systems often lack enforceable spending restrictions, allowing donated funds to be diverted toward unintended purposes.

Misuse is frequently detected only after financial damage has occurred.

7.4 Fragmented Information

Critical information remains distributed across multiple independent systems.

Examples include:

- Donation platforms
- Government databases
- NGO records
- Disaster monitoring systems
- Banking infrastructure
- Audit reports

This fragmentation prevents end-to-end visibility.

7.5 Trust Crisis

Many donors discontinue contributions because they cannot independently verify organizational claims.

Current trust models rely primarily on:

- Reputation
- Marketing
- Organizational size

rather than objective evidence.

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8. Existing System Analysis

Current humanitarian platforms successfully solve isolated problems but fail to provide integrated execution.

Existing Solution	Strength	Major Limitation
Traditional NGOs	Large outreach	Limited transparency

Existing Solution	Strength	Major Limitation
Crowdfunding Platforms	Easy fundraising	No impact verification
Government Relief Systems	Large-scale distribution	Bureaucratic delays
Blockchain Projects	Immutable records	Lack operational workflow
AI Solutions	Prediction & analytics	No financial execution

No existing approach combines governance, automation, verification, auditing, and programmable finance into one complete system.

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9. Research Gap

Despite rapid advancements in Artificial Intelligence, Blockchain, Financial Technology, and Digital Governance, there remains no comprehensive humanitarian platform capable of autonomously executing aid distribution while maintaining continuous transparency, immutable accountability, and policy-driven financial control.

Existing systems address isolated components of the humanitarian process, but none integrate the complete execution lifecycle into a unified infrastructure.

AidFlow AI addresses this missing execution layer by combining intelligent decision-making, programmable financial controls, blockchain-backed auditing, and evidence-based trust within one cohesive platform.

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10. Problem Statement

How can humanitarian funds—including donations and disaster relief—be allocated, verified, distributed, monitored, and audited without relying on slow, manual, opaque, and trust-dependent workflows while preserving transparency, fairness, security, privacy, and accountability?

AidFlow AI addresses this challenge by transforming humanitarian intent into programmable, verifiable, and autonomous financial execution.

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11. Objectives

The primary objectives of AidFlow AI include:

- Establish transparent donation ecosystems
- Eliminate blind trust through evidence-based auditing
- Automate verification using Artificial Intelligence
- Prevent misuse through programmable financial controls
- Enable immutable blockchain-backed audit trails
- Improve beneficiary verification
- Strengthen NGO accountability
- Support governments through real-time oversight
- Enhance disaster response efficiency
- Build dynamic trust scores based on measurable behavior
- Reduce fraud and duplicate claims
- Simplify humanitarian governance through intelligent automation

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12. Scope of the Project

AidFlow AI focuses on creating an integrated humanitarian execution platform supporting the complete lifecycle of donations and disaster relief.

The platform includes:

- Donation Management
- Campaign Management
- NGO Operations
- Beneficiary Verification
- Merchant Ecosystem
- Government Monitoring
- AI Verification
- Fraud Detection
- Blockchain Auditing
- Dynamic Trust Scoring
- Public Transparency Portal
- Disaster Relief Workflows
- Queue-Based Background Processing
- Analytics & Reporting

The system is designed to remain modular, scalable, and extensible for future integrations including stablecoins, zero-knowledge proofs, decentralized governance, predictive disaster response, and international humanitarian collaboration.

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13. Stakeholders

AidFlow AI serves multiple stakeholders, each interacting with different parts of the ecosystem.

Stakeholder	Primary Objective
Donor	Transparent and trustworthy donations
NGO	Campaign execution and beneficiary support
Beneficiary	Receive verified assistance
Merchant	Accept controlled relief payments
Government	Policy enforcement and oversight
Administrator	Platform governance
Public	Independent audit verification

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14. Proposed Solution

AidFlow AI introduces a unified humanitarian execution infrastructure that integrates Artificial Intelligence, Blockchain Technology, Rule-Based Automation, and Dynamic Trust Scoring into a single ecosystem.

Human stakeholders define policies and governance rules.

Artificial Intelligence interprets and executes workflows.

Programmable financial controls enforce spending restrictions.

Blockchain permanently records critical audit events.

Dynamic trust scores continuously evolve based on measurable actions rather than organizational reputation.

Together, these components establish a transparent, secure, scalable, and evidence-driven humanitarian ecosystem capable of supporting donations, disaster relief, and future digital governance initiatives.

SECTION B — SYSTEM ARCHITECTURE

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15. System Architecture

AidFlow AI is designed as a modular, service-oriented humanitarian execution infrastructure where each component is responsible for a well-defined domain of functionality. Instead of relying on a monolithic application, the system separates responsibilities into multiple independent modules that communicate through APIs, background workers, event queues, and blockchain audit services.

The architecture prioritizes:

- Scalability
- Transparency
- Fault Tolerance
- Security
- Extensibility
- Maintainability
- Auditability

Every layer has a single responsibility while remaining loosely coupled with other layers.

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16. High-Level Architecture

The complete AidFlow AI ecosystem consists of the following major layers:

```
                         Users
 ------------------------------------------------------------
  Donor | NGO | Beneficiary | Merchant | Government | Admin
 ------------------------------------------------------------
                      React Frontend
 ------------------------------------------------------------
                     REST API Gateway
 ------------------------------------------------------------
                  Express Backend (Node.js)
 ------------------------------------------------------------
 Business Services | Policy Engine | Workflow Engine | Audit
 ------------------------------------------------------------
  Queue Manager (BullMQ + Redis)
 ------------------------------------------------------------
  AI Agents | Blockchain Service | Notification Service
 ------------------------------------------------------------
```

```
MongoDB | Redis | Ethereum Blockchain
```

Each layer performs an independent responsibility while collaborating with adjacent layers.

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17. Layered Architecture

The platform follows a seven-layer architecture.

Layer 1 — Presentation Layer

Responsible for user interaction.

Technology:

- React
- Tailwind CSS
- Vite

Provides interfaces for:

- Authentication
- Dashboard
- Donations
- Campaigns
- Wallet
- Analytics
- Reports
- Audit Viewer

Each user role receives an independent dashboard customized according to permissions.

Layer 2 — API Layer

Acts as the communication bridge between frontend and backend.

Responsibilities

- Request validation
- Authentication

- Authorization
- Route handling
- Rate limiting
- API versioning
- Error handling

Primary Technologies

- Express.js
- JWT
- Middleware

Layer 3 — Business Logic Layer

This layer contains the complete application logic.

Major responsibilities include

- Campaign Management
- Donation Management
- Wallet Management
- Beneficiary Management
- Merchant Management
- Government Operations
- Policy Enforcement
- Trust Calculation
- Workflow Execution

This layer acts as the brain of the backend.

Layer 4 — AI Layer

Artificial Intelligence is responsible for automated verification and decision support.

Current AI Agents include

Eligibility Agent

Determines whether a beneficiary satisfies all predefined eligibility criteria.

Input

- Identity
- Documents
- Income
- Disaster Status
- Previous Assistance

Output

- Eligible
- Review Required
- Rejected

Fraud Detection Agent

Detects suspicious activities.

Examples

- Duplicate donations
- Fake proof uploads
- Suspicious merchants
- Multiple identities
- Repeated claims

Risk Assessment Agent

Calculates transaction risk.

Risk factors include

- Donation history
- NGO reputation
- Merchant behavior
- Beneficiary history
- Geographic anomalies

Produces

- Low Risk
- Medium Risk
- High Risk

Future AI Modules

Future versions may include

- NLP Policy Interpreter
- Explainable AI
- Disaster Prediction
- Image Verification
- OCR Verification
- Voice Complaint Analysis
- Beneficiary Recommendation Engine

Layer 5 — Background Processing Layer

Not every task should execute immediately during API requests.

Heavy operations are delegated to asynchronous workers.

Technology

- BullMQ
- Redis
- Node Workers

Current workers include

- Donation Worker
- AI Worker
- Fraud Worker
- Blockchain Worker
- Wallet Expiry Worker
- Notification Worker
- Recurring Donation Worker

Benefits

- Faster API response
- Better scalability
- Fault tolerance
- Retry mechanism
- Parallel processing

Layer 6 — Blockchain Layer

AidFlow does not store complete application data on-chain.

Instead, blockchain is used only for integrity verification.

Stored information

- Merkle Root
- Timestamp
- Transaction Hash
- Block Number

Never stored

- Personal Information
- Documents
- Images
- Receipts

Advantages

- Privacy
- Low Cost
- Fast
- Immutable Proof

Layer 7 — Database Layer

AidFlow uses MongoDB as its primary operational database.

Major collections include

- Users
- Campaigns
- Donations
- Wallets
- Beneficiaries
- NGOs
- Merchants
- Policies
- Audit Logs
- Notifications

- Trust Scores

Redis acts as

- Queue storage
- Cache
- Session management
- Worker communication

---

18. Component Architecture

AidFlow consists of multiple independent modules.

```
Frontend

Authentication
Dashboard
Campaign
Donation
Wallet
Analytics
Audit Viewer

↓

Backend

Auth Module
Campaign Module
Donation Module
Wallet Module
Merchant Module
Beneficiary Module
Government Module
Admin Module
Policy Module
Audit Module

↓

AI Services
```

```
Eligibility
Fraud
Risk

↓

Blockchain Service

Hash
Merkle Tree
Anchor

↓

Database
```

Every component can evolve independently without affecting unrelated modules.

---

19. Backend Module Architecture

The backend follows modular architecture.

```
Backend

├── Authentication
├── Authorization
├── Users
├── Campaigns
├── Donations
├── Beneficiaries
├── NGOs
├── Government
├── Merchants
├── Wallets
├── AI
├── Audit
├── Blockchain
├── Notification
├── Reports
├── Analytics
```

```
├── Policies
└── Workers
```

Every module contains

- Routes
- Controller
- Service
- Validation
- Model
- Utilities

This improves maintainability.

---

20. Frontend Architecture

Frontend follows a feature-based architecture.

Major pages include

Authentication

- Login
- Register

Donor

- Dashboard
- Campaigns
- Donations
- Wallet
- Profile

NGO

- Dashboard
- Beneficiaries
- Proof Upload
- Campaign Management

Government

- Dashboard

- Monitoring
- Reports
- Emergency Control

Merchant

- Wallet Scanner
- Transactions
- Settlement

Beneficiary

- Wallet
- Transactions
- History

Admin

- Users
- Policies
- Audit
- AI Monitoring
- System Analytics

---

21. AI Microservice Architecture

The AI subsystem operates independently from the backend.

```
Express Backend

↓

REST API

↓

FastAPI Server

↓

Eligibility Agent
```

```
Fraud Agent

Risk Agent

↓

Prediction

↓

Response

↓

Backend Updates Database
```

Benefits

- Independent scaling
- Better maintainability
- Language independence
- Easier model deployment

---

22. Blockchain Architecture

AidFlow minimizes blockchain usage by anchoring only cryptographic proofs.

Workflow

```
Donation

↓

Audit Event

↓

SHA-256 Hash

↓
```

```
Merkle Tree

↓

Merkle Root

↓

Ethereum Smart Contract

↓

Transaction Hash

↓

Blockchain Explorer
```

Advantages

- Very low gas cost
- Privacy preserving
- Immutable verification
- Public auditability

---

23. Queue Processing Architecture

AidFlow processes heavy jobs asynchronously.

```
API Request

↓

Database Save

↓

Queue Job Created

↓
```

```
Redis

↓

Worker

↓

AI

↓

Blockchain

↓

Notification

↓

Audit Updated
```

Without queues, API response times would increase significantly.

---

24. Request Lifecycle

Example

Donor creates donation.

Step 1

Frontend sends request.

↓

Step 2

Backend validates authentication.

↓

Step 3

Donation stored in MongoDB.

↓

Step 4

Audit entry created.

↓

Step 5

AI verification job queued.

↓

Step 6

Worker processes AI.

↓

Step 7

Risk score generated.

↓

Step 8

Blockchain worker anchors Merkle Root.

↓

Step 9

Trust score updated.

↓

Step 10

Dashboard refreshed.

Entire lifecycle remains auditable.

---

25. Fault Tolerance

AidFlow is designed to continue operating even if certain services become unavailable.

Failure	System Response
AI Offline	Queue request and retry
Blockchain Offline	Delay anchoring
Redis Restart	Recover queued jobs
Worker Crash	Retry automatically
Database Failure	Reject transaction safely
API Restart	Stateless recovery

---

26. Scalability Strategy

AidFlow has been designed for horizontal scalability.

Scalable components include

- Frontend
- Backend APIs
- Workers
- AI Services
- Redis
- MongoDB Replica Sets

Future cloud deployment supports

- Kubernetes
- Docker Swarm
- AWS
- Azure
- GCP

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27. Architectural Principles

AidFlow follows several engineering principles.

Separation of Concerns

Every module performs one responsibility.

Modular Design

Independent services can evolve separately.

Event-Driven Processing

Heavy operations execute asynchronously.

Security by Design

Authentication, authorization, and audit are integrated into every workflow.

Transparency by Default

Every important event creates an immutable audit record.

Privacy Preservation

Personal data never enters the blockchain.

AI-Assisted, Human-Governed

Artificial Intelligence accelerates decision-making while governance policies remain controlled by authorized stakeholders.

End of Section B

The following section explains every user role in detail, including permissions, dashboards, workflows, features, responsibilities, edge cases, and complete business interactions.

SECTION C — USER ROLES & SYSTEM ACTORS

AidFlow AI is a multi-stakeholder humanitarian platform where every participant has clearly defined responsibilities, permissions, workflows, and system interactions.

Unlike traditional donation platforms where every user performs nearly identical operations, AidFlow AI implements Role-Based Access Control (RBAC), ensuring every stakeholder only accesses resources relevant to their responsibilities.

Each role contributes toward one common objective:

> **Deliver humanitarian aid transparently, securely, and accountably.**

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28. User Role Overview

The platform currently supports six primary user roles.

Role	Primary Responsibility
Donor	Contribute funds and monitor their impact
NGO	Manage campaigns, beneficiaries, and proof of utilization
Beneficiary	Receive and utilize approved assistance
Merchant	Accept authorized relief payments
Government	Policy creation, monitoring, and oversight
Administrator	Platform governance and operational management

Each role has independent dashboards, permissions, workflows, and analytics.

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29. Donor Module

Overview

The Donor is the financial contributor of the humanitarian ecosystem.

Unlike conventional donation platforms where the donor loses visibility immediately after making a contribution, AidFlow AI keeps donors connected throughout the complete donation lifecycle.

Every donation remains traceable until final utilization.

The donor becomes an active participant rather than a passive contributor.

Objectives

The donor aims to:

- Support verified humanitarian campaigns
- Ensure transparent utilization of donations
- Monitor impact in real time
- Reduce fraud risks
- Verify proof of work
- Build trust through evidence

Responsibilities

The donor is responsible for:

- Maintaining authentic identity
- Selecting campaigns responsibly
- Making legitimate donations
- Reviewing utilization reports
- Reporting suspicious activities
- Participating in platform feedback

Permissions

The donor can:

✅ Register

✅ Login

✅ Edit profile

✅ View verified NGOs

✅ Browse campaigns

✅ Donate

✅ Track donations

✅ View audit timeline

✅ Download receipts

✅ View blockchain proof

✅ Receive notifications

The donor cannot:

❌ Approve campaigns

❌ Modify beneficiaries

❌ Change policies

❌ Edit trust scores

❌ Access confidential NGO information

Donor Dashboard

The dashboard provides a centralized overview of all donor activities.

Major widgets include

• Donation Summary

• Active Campaigns

• Recent Donations

• Impact Timeline

• Audit History

• Blockchain Verification

• NGO Trust Scores

• Recommended Campaigns

• Notification Center

• Account Overview

Donor Features

Campaign Discovery

Browse verified campaigns using filters such as:

- Disaster Type
- NGO
- Location
- Urgency
- Goal Amount
- Category

Secure Donations

Supports transparent donation flow.

Every donation receives

- Unique Donation ID
- Timestamp
- Audit Record
- Status
- Blockchain Verification Reference

Donation Tracking

Every contribution can be monitored through multiple stages.

Possible statuses

- Created
- Processing
- Approved
- Allocated
- Distributed
- Utilized
- Verified
- Completed

Public Audit Viewer

The donor may inspect

- Transaction Timeline
- NGO Updates
- Uploaded Proofs
- Blockchain Hash
- Merkle Verification
- AI Verification Status

NGO Trust Scores

AidFlow calculates trust dynamically.

Factors include

- Historical performance
- Verification accuracy
- Fraud incidents
- Campaign completion rate
- Proof quality
- Community feedback

Notification System

Examples

Donation received

Campaign completed

Proof uploaded

Aid distributed

Blockchain anchored

Trust score updated

Emergency campaign launched

Donor Workflow

Step 1

Register account

↓

Step 2

Verify identity

↓

Step 3

Browse campaigns

↓

Step 4

Select NGO

↓

Step 5

Donate

↓

Step 6

Donation recorded

↓

Step 7

Audit created

↓

Step 8

Blockchain proof generated

↓

Step 9

Track utilization

↓

Step 10

Verify impact

↓

Step 11

Receive completion report

Donor Business Flow

```
Campaign Discovery

↓

Donation

↓

Payment

↓

Audit Record

↓

NGO Allocation

↓

Beneficiary Distribution

↓

Proof Upload

↓

AI Verification

↓

Blockchain Anchor

↓

Impact Report

↓
```

```
Donation Completed
```

Edge Cases

If payment fails

→ Donation cancelled

If NGO rejected

→ Refund initiated

If fraud detected

→ Investigation starts

If blockchain unavailable

→ Queue anchoring

If proof rejected

→ NGO resubmission required

Future Enhancements

- Recurring donations
- Monthly subscriptions
- AI campaign recommendation
- Donation portfolio
- Tax reports
- CSR integration
- International donations

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30. NGO Module

Overview

NGOs form the operational backbone of AidFlow AI.

They create campaigns, verify beneficiaries, distribute aid, upload proofs, and maintain accountability.

Unlike existing platforms where NGOs simply receive funds, AidFlow continuously evaluates operational transparency.

Objectives

The NGO aims to

- Raise funds
- Execute campaigns
- Support beneficiaries
- Upload utilization proofs
- Improve trust score
- Maintain compliance

Responsibilities

The NGO is responsible for

- Campaign planning
- Beneficiary onboarding
- Fund utilization
- Proof submission
- Compliance
- Fraud reporting
- Resource allocation

Permissions

NGOs may

✅ Create campaigns

✅ Manage campaigns

✅ Add beneficiaries

✅ Verify beneficiary documents

✅ Approve distributions

✅ Upload bills

✅ Upload images

✅ Upload geo-tagged proofs

✅ View reports

✅ Monitor trust score

NGOs cannot

❌ Change platform policies

❌ Modify blockchain records

❌ Edit audit logs

❌ Access other NGO data

NGO Dashboard

Major sections

Campaign Analytics

Beneficiary Management

Donation Statistics

Fund Utilization

Pending Proofs

Trust Score

Audit Timeline

Notifications

Reports

Wallet Overview

AI Alerts

Campaign Lifecycle

Campaign Creation

↓

Verification

↓

Published

↓

Receives Donations

↓

Funds Allocated

↓

Beneficiary Distribution

↓

Proof Upload

↓

AI Verification

↓

Audit Completed

↓

Campaign Closed

Beneficiary Management

NGOs maintain beneficiary records.

Information includes

- Identity
- Address
- Family Details
- Verification Status
- Aid Category
- Distribution History
- Wallet Status

Proof Upload

NGOs upload

Bills

Invoices

Receipts

Geo-tagged Photos

Videos

Documents

Completion Reports

These proofs become inputs for AI verification.

AI Validation

Uploaded proofs undergo

Duplicate Detection

Metadata Validation

Location Verification

Timestamp Validation

Forgery Detection

Consistency Checks

Risk Assessment

Trust Score Calculation

NGO trust increases when

✔ Proofs accepted

✔ Campaigns completed

✔ No fraud detected

✔ Timely execution

✔ Positive beneficiary feedback

Trust decreases when

✖ Fake proofs

✖ Delayed execution

✖ Policy violations

✖ Duplicate submissions

✖ Fraud detection

NGO Workflow

Register

↓

Verification

↓

Create Campaign

↓

Receive Donations

↓

Allocate Funds

↓

Verify Beneficiaries

↓

Distribute Aid

↓

Upload Proof

↓

AI Verification

↓

Blockchain Audit

↓

Campaign Completion

Failure Scenarios

Missing proofs

↓

Reminder generated

↓

Deadline exceeded

↓

Campaign flagged

↓

Government notified

↓

Audit investigation

Future Enhancements

Predictive campaign planning

Inventory management

Volunteer management

Disaster logistics

Supply-chain integration

Offline beneficiary verification

Drone-assisted proof collection

Satellite disaster validation

Donor–NGO Interaction Model

```
Donor
↓
Campaign
↓
Donation
↓
NGO
↓
Aid Distribution
↓
Proof Upload
↓
AI Verification
↓
Blockchain
↓
Donor Dashboard
↓
Verified Impact
```

Summary

The Donor and NGO modules together establish the foundation of the AidFlow ecosystem.

Donors provide financial support while NGOs execute humanitarian operations. Artificial Intelligence continuously validates execution, Blockchain preserves immutable evidence, and the Trust Engine transforms organizational credibility into measurable, data-driven accountability.

31. Beneficiary Module

Overview

The Beneficiary is the primary recipient of humanitarian assistance within the AidFlow AI ecosystem.

Unlike conventional aid systems where beneficiaries often receive unrestricted cash or depend on manual approvals, AidFlow AI introduces a structured, transparent, and policy-driven approach to aid delivery.

Every beneficiary undergoes verification, eligibility assessment, and controlled fund allocation before receiving assistance. Funds are distributed through programmable relief wallets that enforce predefined spending policies, ensuring donations are utilized only for their intended humanitarian purpose.

The beneficiary remains at the center of the ecosystem while maintaining dignity, privacy, transparency, and accountability.

Objectives

The Beneficiary module aims to:

- Receive verified humanitarian assistance.
- Eliminate delays in aid distribution.
- Ensure fair and unbiased eligibility evaluation.
- Protect beneficiary privacy.
- Prevent duplicate or fraudulent claims.
- Enable transparent utilization of allocated resources.
- Improve disaster response efficiency.

Responsibilities

Beneficiaries are responsible for:

- Providing authentic identity information.
- Completing profile verification.
- Uploading required supporting documents.
- Using allocated funds only for approved purposes.
- Cooperating during verification procedures.
- Reporting issues or misuse.
- Maintaining updated personal information.

Permissions

A Beneficiary can:

✅ Register

✅ Complete KYC

✅ Upload documents

✅ View eligibility status

✅ Receive aid

✅ View wallet balance

✅ View transaction history

✅ Spend at approved merchants

✅ View notifications

✅ Download receipts

✅ Report problems

A Beneficiary cannot:

❌ Create campaigns

❌ Modify aid policies

❌ Change wallet restrictions

❌ Edit trust scores

❌ Approve other beneficiaries

❌ Modify blockchain records

Beneficiary Dashboard

The dashboard provides a complete overview of received assistance.

Widgets include

• Profile Status

• Verification Progress

• Wallet Balance

• Aid History

• Active Assistance

• Spending Categories

• Transaction History

• Notifications

• QR Wallet

• Support Center

Registration Workflow

Beneficiary Registration

↓

Identity Verification

↓

Document Upload

↓

AI Eligibility Check

↓

Manual Review (if required)

↓

Approved

↓

Wallet Creation

↓

Ready to Receive Aid

Beneficiary Verification

Verification combines automated AI validation and administrative review.

Verification includes

- Identity Verification
- Government ID
- Address Validation
- Income Validation
- Family Information
- Disaster Impact Assessment
- Previous Aid History
- Duplicate Detection

Future versions may integrate Digital Identity systems and Zero-Knowledge Proofs for privacy-preserving verification.

Eligibility Evaluation

The Eligibility Engine evaluates applicants using multiple criteria.

Examples include

- Income threshold
- Disaster severity
- Household size
- Medical condition
- Vulnerability score
- Previous assistance
- Geographic location
- Government priority rules

Possible Outcomes

✓ Eligible

✓ Conditionally Eligible

✓ Manual Review Required

✖ Rejected

Beneficiary Wallet

Every approved beneficiary receives a programmable digital relief wallet.

Unlike traditional wallets, AidFlow wallets enforce spending rules through policy-based controls.

Wallet information includes

- Wallet ID
- Balance
- Aid Category
- Expiration Date
- Allowed Merchant Types
- Spending Limits
- Transaction History
- Blockchain Verification Reference

Wallet Features

Supported capabilities

• QR Payments

• Merchant Validation

• Category Restrictions

• Balance Tracking

• Spending Notifications

• Automatic Expiry

• Audit History

• Transaction Verification

Spending Restrictions

Funds may only be used for approved humanitarian categories.

Example categories

✓ Food

✓ Medicine

✓ Shelter

✓ Emergency Supplies

✓ Education

✓ Hygiene

Restricted examples

✖ Alcohol

✖ Luxury Goods

✖ Gambling

✖ Cash Withdrawal

✖ Unauthorized Transfers

All restrictions are enforced automatically by the Policy Engine.

Aid Lifecycle

Beneficiary Approved

↓

Wallet Created

↓

Funds Allocated

↓

Notification Sent

↓

Merchant Purchase

↓

Transaction Verified

↓

Audit Generated

↓

Blockchain Anchor

↓

Wallet Updated

↓

Aid Completed

Notifications

Examples

Aid Approved

Funds Received

Wallet Activated

Transaction Successful

Wallet Expiry Reminder

Policy Updates

Merchant Accepted

Proof Verified

Edge Cases

Wallet Expired

↓

Funds returned to campaign

Merchant Offline

↓

Retry transaction

Duplicate Registration

↓

AI flags account

↓

Manual investigation

Fraud Detected

↓

Wallet frozen

↓

Government notified

↓

Audit initiated

Future Enhancements

Biometric Authentication

Offline Wallet

Multi-language Support

Voice Assistant

SMS Notifications

NFC Payments

International Humanitarian Wallet

Stablecoin Support

---

32. Merchant Module

Overview

Merchants form the final execution point of humanitarian assistance.

Instead of allowing unrestricted cash withdrawals, AidFlow AI authorizes only verified merchants to accept humanitarian wallet payments.

Merchants ensure that donated funds are converted into actual goods and services while maintaining complete transaction transparency.

Objectives

Merchant module objectives include

- Accept humanitarian payments.
- Prevent misuse of aid.
- Validate beneficiary purchases.
- Support transparent settlements.
- Improve accountability.
- Reduce fraud.

Responsibilities

Merchants are responsible for

- Maintaining valid registration.
- Selling approved products.
- Following category restrictions.
- Recording transactions.
- Cooperating during audits.
- Maintaining inventory accuracy.

Merchant Permissions

Merchants can

✅ Register

✅ Verify business

✅ Accept wallet payments

✅ Scan QR codes

✅ View transactions

✅ View settlements

✅ Download reports

✅ Receive notifications

Merchants cannot

❌ Modify wallet rules

❌ Change aid policies

❌ Allocate funds

❌ Edit beneficiary records

❌ Modify blockchain records

Merchant Dashboard

Dashboard sections include

Business Overview

Settlement Summary

Daily Transactions

Pending Settlements

Wallet Payments

Inventory Categories

Audit History

Notifications

Support

Analytics

Merchant Registration

Merchant Registration

↓

Business Verification

↓

Category Approval

↓

Document Review

↓

Wallet Integration

↓

Approved Merchant

Payment Flow

Beneficiary selects products

↓

Merchant generates invoice

↓

QR Code scanned

↓

Wallet validates merchant

↓

Category validated

↓

Policy Engine approval

↓

Transaction approved

↓

Merchant receives confirmation

↓

Audit generated

↓

Blockchain anchored

Merchant Validation

Before payment approval the system verifies

- Merchant identity
- Business license
- Approved category
- Wallet eligibility
- Product category
- Spending policy
- Daily transaction limits

Settlement Process

Successful payment

↓

Settlement generated

↓

Audit entry

↓

Blockchain proof

↓

Settlement completed

↓

Reports updated

Merchant Categories

Examples

Food Stores

Medical Shops

Shelter Providers

Educational Suppliers

Emergency Equipment

Water Supply

Essential Goods

Each merchant belongs to one or more approved categories.

Fraud Prevention

AI continuously monitors

- Duplicate invoices
- Fake merchants
- Unusual sales volume
- Suspicious pricing
- Collusion patterns
- Geographic anomalies
- Multiple QR attempts
- Repeated transactions

Suspicious merchants are automatically flagged for review.

Merchant Notifications

Examples

Payment Received

Settlement Completed

Policy Updated

Audit Requested

Inventory Alert

Verification Expiring

System Maintenance

Edge Cases

Merchant Suspended

↓

Payments blocked

Invalid Product Category

↓

Transaction rejected

Wallet Expired

↓

Purchase cancelled

Merchant Verification Expired

↓

Settlement paused

↓

Renewal required

Future Enhancements

POS Integration

Inventory APIs

GST Integration

Offline QR Payments

Supply Chain Integration

IoT Inventory Tracking

AI Demand Forecasting

Cross-border Settlements

Beneficiary–Merchant Interaction

```
Beneficiary

↓

Select Goods

↓

Merchant

↓

Generate Invoice

↓
```

```
QR Payment

↓

Policy Validation

↓

Wallet Verification

↓

Payment Approved

↓

Audit Generated

↓

Blockchain Anchor

↓

Settlement Completed
```

Wallet Ecosystem

```
Government / NGO

↓

Aid Allocation

↓

Beneficiary Wallet

↓
```

```
Approved Merchant

↓

Essential Goods

↓

Audit Trail

↓

Blockchain

↓

Public Verification
```

Summary

The Beneficiary and Merchant modules together represent the execution layer of AidFlow AI.

Beneficiaries receive policy-governed humanitarian assistance through programmable wallets, while verified merchants ensure that aid is converted into essential goods and services. Every transaction is validated through the Policy Engine, monitored by AI, recorded in the Audit Layer, and anchored on the blockchain, creating an end-to-end transparent and accountable aid distribution ecosystem.

33. Government Module

Overview

Government agencies play a strategic governance role within the AidFlow AI ecosystem.

Rather than manually approving every beneficiary or transaction, governments establish policies, monitor humanitarian programs, verify participating organizations, oversee disaster response operations, and intervene only when exceptional circumstances require administrative authority.

AidFlow AI shifts governments from operational execution to policy-driven governance.

Instead of controlling individual transactions, governments define transparent rules that are automatically enforced across the platform.

Objectives

The Government Module aims to

- Create transparent humanitarian policies.
- Monitor NGO compliance.
- Improve disaster response.
- Reduce corruption.
- Enable evidence-based governance.
- Maintain regulatory oversight.
- Support national disaster management.

Responsibilities

Government authorities are responsible for

- Creating humanitarian policies.
- Defining beneficiary eligibility rules.
- Monitoring national campaigns.
- Verifying participating NGOs.
- Managing emergency declarations.
- Reviewing fraud investigations.
- Supervising audit reports.
- Publishing humanitarian statistics.
- Coordinating disaster response.

Permissions

Government users may

✅ Create Policies

✅ Update Policies

✅ Verify NGOs

✅ View National Analytics

✅ Monitor Donations

✅ Monitor Beneficiaries

✅ View AI Reports

✅ Review Fraud Cases

✅ Trigger Emergency Policies

✅ Freeze Campaigns

✅ Export Reports

Government users cannot

❌ Edit blockchain records

❌ Modify audit history

❌ Alter donation records

❌ Directly change beneficiary wallets

❌ Modify AI decisions without audit trail

Government Dashboard

The Government Dashboard provides a national overview of humanitarian operations.

Major components include

• National Dashboard

• Disaster Monitoring

• NGO Registry

• Campaign Statistics

• Beneficiary Statistics

• Fraud Dashboard

• AI Decision Reports

• Emergency Controls

• Audit Explorer

• Geographic Heatmaps

• Analytics

• Policy Manager

Policy Management

Policies define how AidFlow AI behaves.

Policies may include

Eligibility Rules

Donation Rules

Wallet Rules

Spending Restrictions

Merchant Categories

Emergency Response Rules

Verification Thresholds

Trust Score Weights

Fraud Thresholds

Policy versions remain immutable once activated.

Each update creates a new version to preserve historical governance.

Disaster Management

Government agencies can declare

Flood

Earthquake

Cyclone

Wildfire

Pandemic

Conflict

Emergency Food Crisis

Once activated,

AidFlow automatically

- Enables emergency workflows
- Prioritizes beneficiaries
- Adjusts allocation formulas
- Updates AI decision parameters
- Increases monitoring

NGO Verification

Government authorities verify

NGO Registration

Legal Status

Licenses

Tax Information

Operational History

Compliance Records

Previous Fraud

Trust Score

Only verified NGOs can receive humanitarian funds.

National Monitoring

Government agencies receive real-time insights including

Total Donations

Total Campaigns

Beneficiaries Served

Funds Distributed

Pending Verification

Fraud Cases

Emergency Campaigns

Trust Distribution

Geographical Coverage

Response Time

Fraud Investigation

Fraud cases may originate from

AI Detection

Citizen Reports

NGO Complaints

Government Audits

Merchant Alerts

Duplicate Beneficiaries

Investigation Workflow

Fraud Alert

↓

Evidence Collection

↓

Audit Review

↓

Manual Investigation

↓

Decision

↓

Resolution

↓

Trust Score Updated

↓

Public Report

Emergency Response Workflow

Disaster Declared

↓

Government Policy Activated

↓

Priority Beneficiaries Identified

↓

NGOs Assigned

↓

Aid Allocated

↓

Wallets Funded

↓

Merchants Activated

↓

Distribution Completed

↓

Proof Uploaded

↓

AI Verified

↓

Blockchain Audit

↓

Government Dashboard Updated

Reports

Government users may export

Monthly Reports

Disaster Reports

NGO Reports

Beneficiary Reports

Donation Reports

Fraud Reports

Compliance Reports

Audit Reports

Financial Reports

Impact Reports

Notifications

Examples

Emergency Activated

Fraud Detected

NGO Suspended

Campaign Flagged

Policy Updated

Audit Completed

Blockchain Anchored

High-Risk Transaction

Edge Cases

Policy Conflict

↓

Validation Required

↓

Previous Version Active

NGO Fraud

↓

Campaign Suspended

↓

Investigation Started

↓

Beneficiaries Protected

Large Disaster

↓

Emergency Scaling

↓

Priority Allocation

↓

Additional Monitoring

Future Enhancements

Satellite Integration

National Disaster APIs

Weather APIs

International Aid Coordination

Cross-border Relief

Predictive Disaster Response

Digital Twin Simulation

Global Humanitarian Dashboard

---

34. Administrator Module

Overview

Administrators are responsible for managing the operational health of the AidFlow AI platform.

Unlike governments, administrators do not define humanitarian policies.

Instead, they maintain platform infrastructure, user management, AI monitoring, audit integrity, and operational continuity.

Administrators ensure that every component of the platform remains secure, available, and compliant.

Objectives

The Administrator Module aims to

- Maintain platform stability.
- Manage users.
- Configure platform settings.
- Supervise AI services.
- Monitor blockchain services.
- Ensure audit integrity.
- Manage security incidents.

Responsibilities

Administrators manage

Platform Users

Roles

Permissions

Configurations

Audit Logs

Workers

Queues

AI Services

Notifications

System Health

Security

Maintenance

Permissions

Administrators can

✅ Manage Users

✅ Assign Roles

✅ Suspend Accounts

✅ Configure System

✅ View All Dashboards

✅ Restart Workers

✅ Monitor Queues

✅ Manage AI

✅ View Logs

✅ Configure Notifications

✅ Export Data

Administrators cannot

❌ Modify blockchain history

❌ Alter audit logs

❌ Change completed donations

❌ Manipulate trust calculations

Administrator Dashboard

Primary sections

System Health

Users

Roles

Permissions

Queues

Workers

AI Monitoring

Blockchain Status

Audit Logs

Security Events

Analytics

Configuration

Maintenance

User Management

Administrators manage

Donors

NGOs

Beneficiaries

Merchants

Government Users

Other Administrators

Supported actions

Create

Update

Suspend

Delete

Reset Password

Assign Role

Verify Identity

Deactivate

Queue Monitoring

Administrators monitor

AI Queue

Blockchain Queue

Donation Queue

Fraud Queue

Notification Queue

Wallet Queue

Metrics

Pending Jobs

Running Jobs

Failed Jobs

Retry Count

Processing Time

Worker Status

AI Monitoring

Monitor

Eligibility Agent

Fraud Agent

Risk Agent

Response Times

Prediction Accuracy

Failures

Retry Requests

Model Versions

Future modules

Explainable AI

Model Registry

Model Drift Detection

Blockchain Monitoring

Track

Current Block

Anchored Transactions

Failed Anchors

Gas Usage

Smart Contract Status

Retry Queue

Transaction History

Audit Management

Administrators oversee

Audit Timeline

Audit Integrity

Hash Verification

Proof Validation

Report Generation

Compliance Logs

Tampering Alerts

Security Center

Monitors

Failed Logins

Unauthorized Access

Role Violations

Token Expiration

Brute Force Attempts

API Abuse

Suspicious Activity

Security Alerts

Maintenance Mode

Administrator may

Enable Maintenance

Disable Registrations

Pause Campaign Creation

Restart Workers

Backup Database

Restore Database

Clear Cache

Restart Services

Platform Analytics

System Metrics

Daily Users

API Requests

Worker Throughput

Average Response Time

Donation Volume

Queue Performance

Fraud Trends

Audit Growth

Storage Usage

Administrator Workflow

Monitor Platform

↓

Detect Issue

↓

Investigate

↓

Resolve

↓

Verify

↓

Document

↓

Notify Stakeholders

↓

Close Incident

Future Enhancements

Multi-region Deployment

Kubernetes Dashboard

Auto Scaling

Observability

Central Logging

Disaster Recovery Automation

SIEM Integration

AI Infrastructure Monitoring

Government–Administrator Collaboration

```
Government

↓

Policies

↓

AidFlow Platform

↓

Administrator

↓

Infrastructure

↓

AI Services

↓

Workers

↓

Blockchain

↓

Audit

↓

Reports

↓
```

Governance Model

AidFlow AI separates governance into three independent layers.

Policy Layer

Responsible for

Government

NGOs (limited)

Humanitarian Committees

Defines

- Rules
- Eligibility
- Priorities

Execution Layer

Responsible for

Backend

AI Agents

Workers

Wallet Engine

Notification Service

Executes

- Donations
- Verification

- Distribution
- Auditing

Trust Layer

Responsible for

Blockchain

Audit Logs

Merkle Trees

Trust Engine

Ensures

- Transparency
- Accountability
- Immutability
- Public Verification

Summary

The Government and Administrator modules provide the governance backbone of AidFlow AI.

Governments establish humanitarian policies, oversee national operations, and coordinate disaster response, while Administrators ensure the platform remains secure, scalable, and operational.

By separating governance, execution, and trust into independent layers, AidFlow AI minimizes centralized control, improves transparency, and creates a resilient humanitarian infrastructure capable of supporting large-scale donation and disaster relief ecosystems.

SECTION D — COMPLETE FEATURE DOCUMENTATION

This section explains every functional component of AidFlow AI in detail.

Each feature is documented from both business and technical perspectives, including its purpose, workflow, system behavior, validation rules, security considerations, and future enhancements.

The objective of this section is to provide a complete functional specification for every major module of the platform.

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35. Authentication Module

Overview

Authentication is the entry point of the AidFlow AI platform.

It ensures that every user accessing the platform is properly identified before interacting with humanitarian resources.

Since AidFlow AI manages sensitive financial operations, beneficiary information, and governmental workflows, authentication serves as the first security boundary protecting the ecosystem.

Objectives

The Authentication Module aims to

- Verify user identity.
- Protect unauthorized access.
- Secure humanitarian data.
- Support multiple user roles.
- Maintain session integrity.
- Provide scalable authentication mechanisms.

Supported Users

Authentication supports

- Donor
- NGO
- Beneficiary
- Merchant
- Government

- Administrator

Each role receives different permissions after successful login.

Functional Requirements

The module shall provide

✓ User Registration

✓ Secure Login

✓ Logout

✓ JWT Authentication

✓ Password Reset

✓ Email Verification

✓ Session Validation

✓ Token Refresh

✓ Role Identification

✓ Remember Login

Registration Workflow

User selects role

↓

Enter profile information

↓

Email validation

↓

Password validation

↓

OTP / Email verification

↓

Account created

↓

Profile initialized

↓

Redirect to dashboard

Login Workflow

User enters credentials

↓

Server validates

↓

Password verified

↓

JWT generated

↓

Role identified

↓

Permissions loaded

↓

Dashboard opened

Validation Rules

Password

Minimum length

Uppercase required

Lowercase required

Number required

Special character required

Email

Must be unique

Must be verified

Role

Must exist

Must be active

Security Features

BCrypt Password Hashing

JWT Tokens

Secure Cookies

HTTPS

Session Expiration

Rate Limiting

Account Lock

Password Hashing

Refresh Tokens

Failed Login Handling

Invalid Password

↓

Retry Counter

↓

Maximum Attempts Exceeded

↓

Temporary Account Lock

↓

Email Notification

Future Enhancements

OAuth

Google Login

Microsoft Login

Aadhaar Authentication

Digital Identity

Biometric Login

Face Authentication

Passkeys

---

36. Authorization (RBAC)

Overview

AidFlow AI uses Role-Based Access Control (RBAC) to ensure every user accesses only those resources required for their responsibilities.

Authorization occurs after authentication.

Objectives

Provide secure access control.

Prevent privilege escalation.

Protect sensitive humanitarian information.

Maintain least-privilege principles.

Roles

Donor

NGO

Beneficiary

Merchant

Government

Administrator

Permission Categories

Read

Write

Update

Delete

Approve

Verify

Export

Manage

Monitor

Configure

Example Permission Matrix

Module	Donor	NGO	Gov	Admin
Donate	✓	✖	✖	✖
Campaign	View	Manage	Monitor	Manage
Beneficiary	✖	Manage	Monitor	Manage
Policy	✖	✖	Create	Manage
Users	✖	✖	View	Manage

Future Enhancements

Attribute Based Access Control

Policy Based Access

Temporary Roles

Delegated Access

---

37. User Management

Overview

The User Management Module manages every participant within AidFlow AI.

All users are stored within a centralized identity system while maintaining independent role-specific information.

Functionalities

User Registration

Profile Creation

Role Assignment

Identity Verification

Status Management

Account Suspension

Account Recovery

Profile Update

Activity Tracking

User States

Pending

Verified

Active

Suspended

Rejected

Deleted

Archived

User Lifecycle

Registration

↓

Verification

↓

Approval

↓

Active

↓

Suspension (if required)

↓

Recovery

↓

Archived

---

38. Profile Management

Every user maintains a personalized profile.

Typical profile information

Personal Information

Contact Information

Verification Status

Trust Information

Activity History

Notification Preferences

Security Settings

Language

Theme

Accessibility

---

39. Dashboard System

Each role receives an independent dashboard.

Dashboard characteristics

Real-time updates

Role-specific widgets

Charts

Notifications

Recent Activity

Analytics

Shortcuts

Reports

KPIs

Dashboard Types

Donor Dashboard

NGO Dashboard

Beneficiary Dashboard

Merchant Dashboard

Government Dashboard

Administrator Dashboard

Shared Widgets

Notifications

Announcements

Activity Feed

Profile Summary

Help Center

Support

---

40. Notification System

Overview

The Notification Module provides real-time communication between the platform and users.

Notifications improve transparency by informing stakeholders whenever significant humanitarian events occur.

Supported Channels

In-App

Email

SMS (Future)

Push Notifications

WhatsApp (Future)

Notification Categories

Authentication

Donation

Campaign

Wallet

Government

Emergency

Audit

Fraud

AI

Security

Example Notifications

Donation Successful

Campaign Created

Wallet Activated

Fraud Detected

Audit Completed

Blockchain Anchored

Emergency Declared

Policy Updated

Beneficiary Approved

Merchant Verified

Notification Workflow

System Event

↓

Notification Created

↓

Queue

↓

Worker

↓

Channel Selected

↓

Delivered

↓

User Reads

↓

Archived

Notification Priorities

Low

Medium

High

Critical

Emergency

Future Enhancements

AI Personalized Notifications

Smart Recommendations

Digest Emails

Multilingual Notifications

Offline Notification Sync

PART 4B — DONATION ECOSYSTEM

---

41. Campaign Management Module

Overview

The Campaign Management Module serves as the foundation of AidFlow AI's fundraising ecosystem. It enables verified NGOs to create, manage, monitor, and conclude humanitarian campaigns while maintaining complete transparency throughout the campaign lifecycle.

Unlike conventional crowdfunding platforms where campaign updates are largely manual and trust-based, AidFlow AI integrates campaign management with AI verification, blockchain auditing, beneficiary tracking, and impact reporting.

Each campaign becomes an independently auditable humanitarian project.

Objectives

The Campaign Management Module aims to:

- Enable NGOs to launch verified fundraising campaigns.
- Maintain complete transparency throughout campaign execution.
- Track campaign progress in real time.
- Associate beneficiaries with campaigns.
- Link donations to measurable impact.
- Maintain immutable audit trails.

Functional Requirements

The system shall allow NGOs to:

- Create Campaigns
- Update Campaign Information
- Pause Campaigns
- Resume Campaigns
- Close Campaigns
- Assign Beneficiaries
- Upload Supporting Documents
- Upload Proof of Utilization
- Monitor Campaign Analytics
- Generate Reports

Campaign Information

Every campaign contains:

- Campaign ID
- Campaign Name
- Description
- NGO
- Category
- Target Amount
- Amount Raised
- Beneficiary Count
- Start Date
- End Date
- Current Status
- Verification Status
- Trust Rating
- Audit Reference

Campaign Categories

Supported categories include:

- Disaster Relief
- Food Distribution
- Medical Support
- Child Welfare
- Women's Welfare
- Elderly Support
- Education
- Shelter
- Emergency Assistance
- Community Development

Future categories can be added without affecting the existing architecture.

Campaign States

Campaigns transition through predefined lifecycle stages.

```
Draft

↓

Submitted

↓

Verification

↓

Approved

↓

Published

↓

Receiving Donations

↓

Distribution

↓

Impact Verification

↓

Completed

↓

Archived
```

Business Rules

A campaign:

✓ Must belong to a verified NGO.

✓ Must define a funding goal.

✓ Must specify beneficiary criteria.

✓ Must contain required documentation.

✓ Cannot receive donations before approval.

✓ Cannot be deleted after public publication.

AI Integration

AI continuously analyzes campaigns by evaluating:

- Description quality
- Duplicate campaigns
- Risk indicators
- NGO history
- Fraud probability
- Expected impact

High-risk campaigns are flagged for manual review.

Dashboard Widgets

Campaign Summary

Active Donations

Beneficiaries

Goal Progress

Remaining Days

Trust Score

Pending Proofs

Audit Timeline

Edge Cases

Goal exceeded

↓

Campaign remains active until manually closed.

Campaign cancelled

↓

Future donations disabled.

↓

Existing audit preserved.

NGO suspended

↓

Campaign frozen.

↓

Government notified.

Future Enhancements

Recurring campaigns

Campaign templates

AI campaign recommendations

Cross-border campaigns

Corporate CSR campaigns

Volunteer integration

---

42. Donation Management Module

Overview

The Donation Management Module manages the complete financial lifecycle of every donation.

Every donation remains linked to:

- Donor
- Campaign
- NGO
- Beneficiary
- Wallet
- Audit
- Blockchain Proof

ensuring complete end-to-end traceability.

Objectives

The Donation Module aims to:

- Receive secure donations.
- Track donation utilization.
- Maintain auditability.
- Prevent financial misuse.
- Enable complete transparency.

Functional Requirements

The system shall support

One-time donations

Recurring donations

Anonymous donations

Corporate donations

Campaign donations

Disaster donations

Refund processing

Receipt generation

Donation tracking

Audit verification

Donation Information

Each donation stores

Donation ID

Donor

Campaign

Amount

Payment Method

Timestamp

Status

Transaction ID

Audit ID

Blockchain Reference

Donation Lifecycle

```
Donation Initiated

↓

Payment Success
```

```
↓

Donation Recorded

↓

Audit Generated

↓

Campaign Updated

↓

Queue Created

↓

AI Verification

↓

Blockchain Anchored

↓

Beneficiary Allocation

↓

Completed
```

Donation Status

Pending

Processing

Successful

Allocated

Distributed

Verified

Refunded

Cancelled

Failed

Validation Rules

Minimum donation amount

Valid campaign required

Verified NGO required

Secure payment confirmation

Unique transaction ID

Duplicate prevention

Receipt Generation

Each donation generates

Digital Receipt

Donation Certificate

Audit Reference

Blockchain Reference

Tax Information

QR Verification

Refund Workflow

Refund Requested

↓

Eligibility Check

↓

Approval

↓

Payment Gateway

↓

Refund Completed

↓

Audit Updated

Fraud Prevention

Duplicate payment detection

Transaction anomalies

Repeated refunds

Abnormal donation patterns

Suspicious payment sources

Velocity checks

Future Enhancements

Crypto donations

Stablecoin donations

International payments

UPI AutoPay

Recurring subscriptions

Employer matching

---

43. Beneficiary Management Module

Overview

Beneficiary Management ensures humanitarian assistance reaches eligible recipients through structured verification, allocation, and monitoring.

This module connects NGOs, AI, Governments, and Wallet Systems into one unified verification pipeline.

Functionalities

Registration

Verification

Eligibility

Document Management

Aid History

Wallet Assignment

Trust Monitoring

Case Management

Status Tracking

Beneficiary States

Applied

↓

Pending Verification

↓

Verified

↓

Eligible

↓

Receiving Aid

↓

Completed

↓

Archived

Stored Information

Beneficiary ID

Identity

Address

Family Details

Income

Aid History

Verification Status

Wallet Status

Risk Score

Trust Information

AI Validation

The AI evaluates

Identity consistency

Duplicate applications

Location

Previous claims

Income

Household data

Fraud indicators

Business Rules

One beneficiary cannot receive duplicate assistance for the same campaign unless explicitly allowed by policy.

Every allocation must remain linked to an audit trail.

---

44. Wallet Management Module

Overview

The Wallet Management Module replaces unrestricted cash distribution with programmable humanitarian wallets.

Instead of trusting beneficiaries to spend funds appropriately, AidFlow AI enforces policy-driven spending restrictions directly within the wallet.

Objectives

Ensure purpose-bound aid.

Prevent misuse.

Increase transparency.

Simplify tracking.

Enable programmable finance.

Wallet Features

Wallet Creation

Wallet Activation

Balance Management

Transaction History

QR Payments

Expiry

Category Restrictions

Audit Tracking

Blockchain Verification

Wallet Lifecycle

```
Beneficiary Approved

↓

Wallet Created

↓

Wallet Activated

↓

Funds Loaded

↓

Transactions

↓

Balance Updated

↓
```

```
Wallet Expired

↓

Closed
```

Spending Policies

Allowed

Food

Medicine

Shelter

Education

Essential Goods

Restricted

Alcohol

Luxury Goods

Cash Withdrawal

Entertainment

Unauthorized Transfers

Wallet States

Created

Active

Suspended

Expired

Frozen

Closed

Validation Rules

Merchant verification required.

Category approval required.

Wallet balance sufficient.

Policy compliance mandatory.

Edge Cases

Expired wallet

↓

Funds returned.

Fraud detected

↓

Wallet frozen.

Merchant unavailable

↓

Retry payment.

---

45. Merchant Transaction Module

Overview

Merchant Transactions convert humanitarian funding into verified goods and services.

Every purchase remains linked to

Wallet

Merchant

Campaign

Donation

Audit

Blockchain

Workflow

```
Beneficiary

↓

Merchant

↓

Invoice

↓

QR Payment

↓

Policy Validation

↓

Wallet Validation

↓

Merchant Validation

↓

Payment Approved

↓
```

```
Settlement

↓

Audit

↓

Blockchain
```

Validation

Merchant verified

Category approved

Wallet active

Funds available

Policy satisfied

QR authentic

Transaction Status

Initiated

Authorized

Completed

Rejected

Refunded

Failed

Cancelled

Settlement

Approved transaction

↓

Settlement created

↓

Merchant notified

↓

Audit updated

↓

Blockchain anchor

↓

Reports generated

Future Enhancements

POS systems

GST integration

Offline payments

Inventory APIs

IoT inventory

Dynamic pricing validation

Business Rules Summary

The Donation Ecosystem follows several mandatory rules:

- Every donation belongs to one campaign.
- Every campaign belongs to one NGO.
- Every beneficiary receives only verified aid.
- Every wallet enforces policy restrictions.

- Every merchant must be verified.
- Every transaction creates an audit event.
- Every audit event contributes to blockchain integrity.
- Every completed campaign generates measurable impact reports.

These rules collectively ensure transparency, accountability, and end-to-end traceability across the entire humanitarian lifecycle.

End of Part 4B

This section documents the complete donation ecosystem, from campaign creation through donation collection, beneficiary verification, programmable wallet allocation, merchant transactions, settlement, auditing, and impact tracking.

PART 4C — AI, AUTOMATION & INTELLIGENT EXECUTION

---

46. Artificial Intelligence Layer

Overview

Artificial Intelligence is the intelligence layer of AidFlow AI.

Unlike conventional donation systems where every verification, approval, and validation depends entirely on manual review, AidFlow AI integrates multiple AI agents that continuously analyze, verify, score, monitor, and assist humanitarian workflows.

The AI layer is not intended to replace governments or NGOs. Instead, it augments human decision-making by automating repetitive verification tasks, detecting anomalies, prioritizing cases, and providing explainable recommendations while keeping final governance under human control.

AI operates as an execution accelerator rather than an autonomous authority.

Objectives

The AI Layer aims to

- Reduce manual verification.
- Detect fraudulent activities.
- Improve beneficiary validation.
- Prioritize humanitarian assistance.
- Increase processing speed.
- Reduce operational costs.
- Improve transparency.
- Support explainable decisions.

Design Principles

AidFlow AI follows six fundamental AI principles.

Human Governed

Humans define policies.

AI executes policies.

Explainable

Every important decision should include a clear explanation.

No "black-box" decisions.

Privacy First

Personal information is never unnecessarily exposed.

Fairness

AI recommendations should remain unbiased.

Continuous Learning

Models improve using verified historical data.

Auditability

Every prediction is logged.

Every decision remains reviewable.

AI Architecture

```
External Data

↓

Backend API

↓

AI Gateway

↓

Eligibility Agent

↓

Fraud Agent

↓

Risk Agent

↓

Decision Engine

↓
```

```
Recommendation

↓

Audit Log

↓

Database
```

AI Components

AidFlow currently consists of three major AI agents.

1. Eligibility Agent
2. Fraud Detection Agent
3. Risk Assessment Agent

Future versions support additional agents.

---

47. Eligibility Engine

Overview

The Eligibility Engine determines whether a beneficiary satisfies humanitarian assistance criteria.

Rather than relying solely on manual verification, AI evaluates multiple factors simultaneously to improve consistency and reduce processing time.

Objectives

- Reduce verification delays.
- Maintain fairness.
- Prevent duplicate beneficiaries.
- Prioritize vulnerable applicants.

- Support policy-driven allocation.

Inputs

Identity

Income

Family Details

Location

Disaster Impact

Medical Information

Government Data

Previous Aid History

Documents

AI Evaluation Criteria

Household Size

Income

Disability

Senior Citizens

Children

Disaster Severity

Employment Status

Existing Assistance

Duplicate Detection

Policy Rules

Outputs

Eligible

Conditionally Eligible

Manual Review Required

Rejected

Example Workflow

Application Submitted

↓

Document Validation

↓

Identity Verification

↓

Duplicate Detection

↓

Policy Evaluation

↓

Eligibility Score

↓

Decision

↓

Audit Log

Business Rules

Every eligibility decision must

Follow current policy version.

Generate an audit record.

Remain reviewable.

Support manual override.

Future Improvements

OCR

Facial Verification

Government APIs

Satellite Verification

Digital Identity

Zero Knowledge Proofs

---

48. Fraud Detection Engine

Overview

Fraud Detection continuously analyzes platform activity to identify suspicious behaviour before financial damage occurs.

Unlike traditional systems where fraud is discovered after audits, AidFlow AI attempts to detect suspicious activities in real time.

Objectives

Reduce fraud.

Protect donations.

Protect beneficiaries.

Improve NGO credibility.

Reduce financial leakage.

Fraud Categories

Duplicate Beneficiary

Fake NGO

Duplicate Documents

Fake Proof Upload

Repeated Claims

Wallet Abuse

Merchant Collusion

Campaign Manipulation

Money Laundering Indicators

Identity Fraud

AI Inputs

Transaction History

Wallet Usage

Documents

Metadata

Location

Time

Merchant Activity

Donation Behaviour

NGO Performance

Audit History

Example Workflow

New Transaction

↓

Feature Extraction

↓

Fraud Model

↓

Risk Score

↓

Threshold Comparison

↓

Flag

↓

Audit Created

↓

Manual Review

Fraud Risk Levels

Low

Medium

High

Critical

Example Rules

Same receipt uploaded twice

↓

High Risk

Same beneficiary appears in multiple campaigns

↓

Investigation Required

Merchant repeatedly sells prohibited products

↓

Merchant Suspended

Future Enhancements

Graph Neural Networks

Behavior Analytics

Device Fingerprinting

Network Analysis

Anomaly Detection

---

49. Risk Assessment Engine

Overview

The Risk Engine predicts operational risk before humanitarian actions are executed.

Rather than waiting for failures, AidFlow AI proactively evaluates potential threats.

Risk Factors

NGO Trust

Campaign History

Beneficiary Risk

Merchant Risk

Geographic Risk

Disaster Severity

Historical Fraud

Policy Compliance

Outputs

Low Risk

Medium Risk

High Risk

Critical Risk

Uses

Campaign Approval

Donation Monitoring

Beneficiary Verification

Merchant Validation

Government Alerts

Priority Allocation

---

50. AI Decision Engine

The Decision Engine combines outputs from all AI agents.

```
Eligibility

+

Fraud

+

Risk

↓

Decision Engine

↓

Recommendation

↓

Backend

↓

Audit
```

Possible outcomes

Approve

Reject

Manual Review

Escalate

Delay

Monitor

Decision Matrix

Eligibility	Fraud	Risk	Decision
High	Low	Low	Approve
High	High	Medium	Manual Review
Medium	Low	High	Escalate
Low	High	Critical	Reject

Explainable AI (XAI)

Every recommendation contains

Decision

Confidence Score

Contributing Factors

Policy Version

Timestamp

Supporting Evidence

Review Recommendation

---

51. Policy Engine

Overview

The Policy Engine acts as the rule interpreter of AidFlow AI.

Policies define

Eligibility

Wallet Rules

Aid Categories

Donation Limits

Merchant Categories

Fraud Thresholds

Trust Weights

Emergency Rules

Objectives

Ensure consistent execution.

Separate policy from implementation.

Support future policy updates.

Enable versioning.

Policy Lifecycle

Draft

↓

Review

↓

Approval

↓

Activation

↓

Execution

↓

Archived

Policy Versioning

Policies cannot be edited after activation.

Instead,

Version 1

↓

Version 2

↓

Version 3

History remains preserved.

---

52. Workflow Engine

Overview

The Workflow Engine orchestrates every business process inside AidFlow AI.

Instead of hardcoding logic across services, workflows define

Actions

Conditions

Triggers

Escalations

Retries

Notifications

Example Workflow

Donation Created

↓

Audit Entry

↓

Queue

↓

AI

↓

Risk

↓

Blockchain

↓

Notification

↓

Dashboard Update

Supported Workflows

Donation

Campaign

Beneficiary

Wallet

Merchant

Fraud

Audit

Government

Emergency

Blockchain

Notifications

Workflow Characteristics

Event Driven

Asynchronous

Fault Tolerant

Retry Support

Scalable

Auditable

---

53. Queue Management

AidFlow processes long-running operations asynchronously.

Technology

Redis

BullMQ

Workers

Current Queues

AI Queue

Donation Queue

Fraud Queue

Blockchain Queue

Wallet Queue

Notification Queue

Audit Queue

Benefits

Fast API

Background Processing

Retry

Parallel Execution

Fault Isolation

Improved User Experience

Queue Workflow

```
API

↓

Job

↓

Redis

↓

Worker

↓

Process

↓
```

```
Database

↓

Audit

↓

Notification
```

---

54. Worker Architecture

Workers execute background tasks.

Current workers

Donation Worker

AI Worker

Fraud Worker

Blockchain Worker

Wallet Worker

Notification Worker

Recurring Donation Worker

Worker Responsibilities

Read queue

Process task

Update database

Generate audit

Notify users

Retry failures

Retry Strategy

Failed Job

↓

Retry 1

↓

Retry 2

↓

Retry 3

↓

Dead Letter Queue

↓

Administrator Alert

---

55. AI Governance

AidFlow AI follows responsible AI principles.

Human Oversight

Critical decisions remain reviewable.

Transparency

Every prediction is logged.

Accountability

AI recommendations never bypass audit.

Privacy

Sensitive information remains protected.

Continuous Monitoring

Model performance is continuously evaluated.

Explainability

Users can understand why a recommendation was generated.

Future AI Roadmap

Future AI modules may include

- Disaster Prediction
- Satellite Image Analysis
- NLP Policy Interpreter
- OCR Document Verification
- Face Matching
- Voice Complaint Analysis
- Demand Forecasting
- Volunteer Allocation
- Resource Optimization
- Explainable Recommendation Dashboard

Summary

The AI and Automation Layer transforms AidFlow AI from a traditional donation management platform into an intelligent humanitarian execution infrastructure.

Rather than relying solely on manual approvals, the system combines policy-driven governance, AI-assisted verification, fraud detection, explainable recommendations, event-driven workflows, queue-based processing, and background workers to deliver transparent, scalable, and accountable humanitarian operations.

PART 4D — TRUST, TRANSPARENCY & AUDIT INFRASTRUCTURE

---

56. Blockchain Integration

Overview

Blockchain serves as the immutable trust layer of AidFlow AI.

Unlike blockchain-first applications that store complete datasets on-chain, AidFlow AI follows a **hybrid architecture** where operational data remains in secure databases while cryptographic proofs are anchored onto the blockchain.

This approach achieves:

- Transparency
- Integrity
- Privacy
- Cost Efficiency
- Scalability

without exposing sensitive beneficiary information.

Objectives

The Blockchain Layer aims to:

- Provide immutable auditability.
- Prevent tampering.
- Establish public trust.
- Minimize blockchain storage costs.

- Protect personally identifiable information (PII).

Why Blockchain?

Traditional databases allow administrators to modify records.

Blockchain ensures:

- Data cannot be secretly modified.
- Historical events remain permanent.
- Independent verification is possible.
- Trust becomes evidence-based.

Stored On-Chain

AidFlow stores only cryptographic proofs.

Examples

- Merkle Root
- Timestamp
- Transaction Hash
- Block Number
- Smart Contract Event

Never Stored On-Chain

- Beneficiary Information
- Donor Information
- Identity Documents
- Medical Records
- Financial Documents
- Images
- Receipts
- Proof Files

Blockchain Workflow

```
Application Event

↓

Audit Record

↓

SHA-256 Hash

↓

Merkle Tree

↓

Merkle Root

↓

Smart Contract

↓

Ethereum Network

↓

Transaction Hash

↓

Public Verification
```

Benefits

Tamper Proof

Public Verification

Low Gas Cost

Privacy Preserving

Immutable History

Independent Validation

---

57. Smart Contract Layer

Purpose

The smart contract acts as the immutable anchor for audit verification.

It does **not** execute business logic such as donation approval or beneficiary verification.

Business logic remains within the backend.

The smart contract stores only:

- Merkle Root
- Timestamp
- Event Metadata

Responsibilities

Accept Merkle Root

Store Immutable Record

Emit Blockchain Events

Support Verification

Prevent Duplicate Anchors

Future Improvements

Stablecoin Settlement

DAO Governance

Cross-chain Support

Layer-2 Networks

---

58. Merkle Tree Infrastructure

Overview

AidFlow AI uses Merkle Trees to efficiently verify thousands of audit events with a single blockchain transaction.

Instead of writing every audit event individually, all events generated within a period are hashed together into one Merkle Tree.

Only the Merkle Root is written to the blockchain.

Workflow

```
Audit Event 1

Audit Event 2

Audit Event 3

Audit Event 4

↓

Hash

↓

Merkle Tree

↓

Merkle Root

↓
```

```
Blockchain
```

Advantages

Low Cost

Scalable

Fast Verification

Tamper Detection

Efficient Storage

---

59. Audit Trail

Overview

Every significant system action generates an audit event.

Audit trails establish accountability across every stakeholder.

No important operation occurs without creating a corresponding audit record.

Events Logged

User Login

Registration

Donation

Campaign Creation

Beneficiary Approval

Wallet Creation

Merchant Payment

AI Recommendation

Fraud Alert

Government Action

Policy Change

Blockchain Anchor

Administrator Action

Audit Information

Audit ID

Actor

Role

Timestamp

Action

Entity

Before State

After State

Hash

Verification Status

Audit Lifecycle

```
System Action

↓

Audit Created

↓
```

```
Hash Generated

↓

Database Stored

↓

Merkle Tree

↓

Blockchain Anchor

↓

Verification Available
```

Audit Characteristics

Immutable

Chronological

Tamper Evident

Searchable

Traceable

Verifiable

---

60. Public Audit Portal

Overview

AidFlow AI provides a public verification portal allowing anyone to verify humanitarian transactions without requiring administrative access.

This increases public confidence and enables independent verification.

Supported Searches

Donation ID

Campaign ID

Audit ID

Blockchain Transaction

Merkle Root

Wallet ID

Portal Features

Audit Timeline

Blockchain Explorer Link

Verification Status

Proof History

Campaign Summary

Trust Indicators

Public Verification Workflow

```
Donation ID

↓

Audit Retrieved

↓

Hash Compared

↓
```

```
Blockchain Verified

↓

Result Displayed
```

---

61. Dynamic Trust Engine

Overview

Trust within AidFlow AI is continuously calculated rather than manually assigned.

Every stakeholder accumulates a trust profile based on measurable behaviour.

Trust is dynamic, evidence-based, and continuously updated.

Objectives

Reward honest behaviour.

Identify risky entities.

Improve transparency.

Support intelligent recommendations.

Reduce fraud.

Entities Receiving Trust Scores

NGOs

Beneficiaries

Merchants

Campaigns

AI Models

Distribution Channels

Trust Inputs

Campaign Success

Proof Accuracy

Fraud Incidents

Policy Compliance

Beneficiary Feedback

Audit History

Verification Results

Transaction Behaviour

Response Time

Trust Outputs

Excellent

Good

Moderate

Low

Critical

Example Workflow

Successful Campaign

↓

Verified Proof

↓

Positive Feedback

↓

Trust Increased

↓

Future Recommendation

Fraud Detected

↓

Investigation

↓

Trust Reduced

↓

Monitoring Increased

---

62. Analytics Module

Overview

AidFlow AI provides real-time analytics to support operational visibility and decision making.

Each role receives analytics appropriate to their responsibilities.

Analytics Categories

Donation Analytics

Campaign Analytics

Beneficiary Analytics

NGO Analytics

Government Analytics

Fraud Analytics

Wallet Analytics

Merchant Analytics

AI Analytics

Blockchain Analytics

KPIs

Total Donations

Active Campaigns

Aid Distributed

Average Response Time

Verification Time

Fraud Rate

Trust Score Distribution

Wallet Utilization

Merchant Activity

Beneficiary Coverage

Visualization

Bar Charts

Pie Charts

Line Charts

Heat Maps

Geographical Maps

Tables

Trend Graphs

---

63. Reporting System

Overview

The Reporting Module generates structured reports for all stakeholders.

Report Types

Daily Reports

Weekly Reports

Monthly Reports

Campaign Reports

Donation Reports

Beneficiary Reports

NGO Reports

Government Reports

Audit Reports

Fraud Reports

Impact Reports

Financial Reports

Export Formats

PDF

CSV

Excel

JSON

Scheduled Reports

The platform supports automatic report generation for administrators and government agencies.

---

64. Logging Infrastructure

Overview

Logs capture operational events for debugging, monitoring, and incident analysis.

Unlike audit records, logs are primarily intended for technical troubleshooting.

Log Types

Application Logs

API Logs

Worker Logs

Database Logs

AI Logs

Blockchain Logs

Security Logs

Performance Logs

Log Levels

INFO

WARNING

ERROR

CRITICAL

DEBUG

---

65. Monitoring & Observability

Objectives

Maintain platform health.

Detect failures early.

Improve reliability.

Reduce downtime.

Monitored Components

Backend

Workers

Redis

MongoDB

AI Services

Blockchain Service

Queues

Notification Service

Metrics

CPU Usage

Memory Usage

Queue Length

Worker Status

Average Response Time

Error Rate

Blockchain Anchoring Success

AI Processing Time

---

66. Disaster Recovery

Objectives

Maintain service continuity during failures.

Protect humanitarian data.

Reduce downtime.

Backup Strategy

Database Backup

Audit Backup

Configuration Backup

Blockchain References

Encrypted Storage

Recovery Strategy

Failure Detected

↓

Restore Backup

↓

Restart Services

↓

Verify Integrity

↓

Resume Operations

---

67. Performance Considerations

AidFlow AI has been designed for large-scale humanitarian operations.

Performance objectives include

- Fast API responses
- Background processing
- Horizontal scalability
- Efficient blockchain usage
- Database indexing
- Queue optimization
- Cache utilization

---

68. Scalability Strategy

The architecture supports independent scaling of:

Frontend

Backend

AI Services

Redis

Workers

MongoDB

Blockchain Gateway

Notification Service

Each component can scale horizontally without impacting unrelated services.

---

69. Security Integration

Security is embedded across every feature.

Major protections include

JWT Authentication

Role-Based Access Control

Password Hashing

HTTPS

Rate Limiting

Input Validation

Audit Trails

Blockchain Verification

Encrypted Secrets

Secure Environment Variables

---

70. Feature Dependency Map

```
Authentication
        │
        ▼
Authorization
        │
        ▼
```

```
User Management
        |
        ▼
Campaigns
        |
        ▼
Donations
        |
        ▼
Beneficiary Verification
        |
        ▼
Wallet Allocation
        |
        ▼
Merchant Transactions
        |
        ▼
Audit Trail
        |
        ▼
Merkle Tree
        |
        ▼
Blockchain
        |
        ▼
Public Verification
```

Section D Summary

The Feature Documentation section defines every major functional component of AidFlow AI.

Collectively, these modules establish a secure, scalable, transparent, and intelligent humanitarian infrastructure where every stakeholder interaction is authenticated, every financial transaction is traceable, every operational decision is auditable, and every trust relationship is backed by measurable evidence rather than assumptions.

The modular architecture ensures that each feature can evolve independently while maintaining seamless integration across the complete humanitarian ecosystem.

SECTION E — COMPLETE BUSINESS WORKFLOWS & SYSTEM PROCESSES

---

71. System Workflow Overview

AidFlow AI is an end-to-end humanitarian execution infrastructure where every operation—from donor registration to final beneficiary utilization—is executed through interconnected workflows governed by policies, AI validation, audit logging, and blockchain verification.

Unlike traditional donation platforms where individual modules operate independently, AidFlow AI connects every stakeholder into a unified execution pipeline.

Every workflow follows the same architectural philosophy:

- Authenticate
- Authorize
- Validate
- Execute
- Audit
- Verify
- Notify
- Archive

This ensures consistency across the platform.

---

72. Complete Donation Lifecycle

Objective

Describe the complete lifecycle of a donation from donor contribution to final humanitarian impact.

Workflow

```
Donor Registration
        │
        ▼
Authentication
        │
        ▼
Browse Campaigns
        │
        ▼
Campaign Selection
        │
        ▼
Donation Initiated
        │
        ▼
Payment Gateway
        │
        ▼
Payment Successful
        │
        ▼
Donation Record Created
        │
        ▼
Audit Event Created
        │
        ▼
Queue Job Created
        │
        ▼
AI Risk Evaluation
        │
        ▼
Campaign Updated
        │
        ▼
NGO Receives Funds
        │
        ▼
Beneficiary Allocation
        │
        ▼
Wallet Created
        │
```

```
        ▼
Merchant Payment
        |
        ▼
Proof Uploaded
        |
        ▼
AI Verification
        |
        ▼
Merkle Tree
        |
        ▼
Blockchain Anchor
        |
        ▼
Impact Report
        |
        ▼
Donation Completed
```

Actors

Donor

NGO

Beneficiary

Merchant

AI

Workers

Blockchain

Government

Administrator

Output

Donation Completed

Audit Available

Blockchain Verified

Trust Updated

Impact Visible

---

73. Campaign Lifecycle

Campaign Draft

↓

NGO Submission

↓

Verification

↓

Approval

↓

Published

↓

Donations Received

↓

Beneficiary Allocation

↓

Aid Distribution

↓

Proof Collection

↓

AI Verification

↓

Audit

↓

Campaign Closed

↓

Archived

Campaign States

Draft

Pending Review

Approved

Active

Paused

Completed

Archived

Cancelled

---

74. Beneficiary Verification Workflow

Registration

↓

Identity Upload

↓

Document Validation

↓

AI Duplicate Check

↓

Policy Evaluation

↓

Risk Assessment

↓

Government Review (if needed)

↓

Approved

↓

Wallet Created

Alternative Flow

If AI confidence is low

↓

Manual Review

↓

Decision

↓

Continue

Failure Flow

Invalid Documents

↓

Re-upload Requested

↓

Rejected if unresolved

---

75. Wallet Allocation Workflow

Approved Beneficiary

↓

Wallet Created

↓

Wallet Activated

↓

Aid Allocated

↓

Notification Sent

↓

Wallet Ready

Business Rules

Wallet belongs to one beneficiary.

Wallet follows active policy.

Wallet expires automatically.

Wallet cannot transfer funds.

---

76. Merchant Payment Workflow

Beneficiary Selects Goods

↓

Merchant Creates Invoice

↓

QR Generated

↓

Wallet Validation

↓

Merchant Validation

↓

Category Validation

↓

Policy Validation

↓

Transaction Approved

↓

Settlement

↓

Audit Generated

↓

Blockchain Anchor

Failure Flow

Wallet Expired

↓

Transaction Cancelled

Merchant Suspended

↓

Payment Rejected

Category Restricted

↓

Purchase Denied

---

77. NGO Operational Workflow

NGO Registration

↓

Verification

↓

Campaign Creation

↓

Campaign Approval

↓

Receive Donations

↓

Beneficiary Verification

↓

Aid Distribution

↓

Upload Proof

↓

AI Verification

↓

Audit

↓

Campaign Completed

Failure Flow

Missing Proof

↓

Reminder

↓

Deadline Exceeded

↓

Campaign Flagged

↓

Government Review

---

78. Government Monitoring Workflow

Emergency Declared

↓

Policy Activated

↓

Affected Areas Identified

↓

Eligible Beneficiaries Calculated

↓

NGOs Assigned

↓

Aid Distributed

↓

Monitoring

↓

Fraud Detection

↓

Reports

↓

Audit

---

79. Administrator Workflow

Monitor Platform

↓

Detect Problem

↓

Investigate

↓

Resolve

↓

Restart Services (if required)

↓

Verify

↓

Close Incident

---

80. AI Decision Workflow

Application

↓

Eligibility Engine

↓

Fraud Engine

↓

Risk Engine

↓

Decision Engine

↓

Recommendation

↓

Audit

↓

Backend

↓

Final Decision

Possible Outcomes

Approve

Reject

Manual Review

Escalate

Monitor

Delay

---

81. Fraud Detection Workflow

Event Created

↓

Feature Extraction

↓

Fraud Model

↓

Risk Score

↓

Threshold

↓

Flag

↓

Audit

↓

Manual Review

↓

Resolution

Resolution Types

False Positive

Warning

Suspension

Investigation

Permanent Ban

---

82. Blockchain Workflow

System Event

↓

Audit Created

↓

Hash

↓

Merkle Tree

↓

Merkle Root

↓

Smart Contract

↓

Ethereum

↓

Transaction Hash

↓

Verification Portal

---

83. Audit Workflow

Every important event follows the same audit lifecycle.

Action

↓

Audit Record

↓

Hash

↓

Stored

↓

Queued

↓

Blockchain

↓

Verification

---

84. Notification Workflow

System Event

↓

Notification Created

↓

Priority Assigned

↓

Queue

↓

Worker

↓

Delivery

↓

User Reads

↓

Archive

---

85. Queue Processing Workflow

API Request

↓

Database Save

↓

Queue Created

↓

Redis

↓

Worker

↓

Processing

↓

Database Update

↓

Audit

↓

Notification

Retry Strategy

Failure

↓

Retry

↓

Retry

↓

Retry

↓

Dead Letter Queue

↓

Administrator Alert

---

86. Public Verification Workflow

Donation ID

↓

Audit Record

↓

Hash

↓

Merkle Proof

↓

Blockchain

↓

Verification

↓

Public Result

---

87. Disaster Relief Workflow

Government Declares Disaster

↓

Emergency Policy Activated

↓

Affected Areas Imported

↓

Beneficiaries Prioritized

↓

Aid Allocation

↓

Wallet Funding

↓

Merchant Network Activated

↓

Aid Utilization

↓

Proof Upload

↓

AI Verification

↓

Blockchain

↓

Government Report

---

88. Trust Score Workflow

System Event

↓

Performance Evaluated

↓

Trust Inputs Calculated

↓

Trust Engine

↓

Score Updated

↓

Recommendations Updated

↓

Dashboard Refreshed

---

89. Cross-Role Interaction Matrix

Actor	Interacts With
Donor	NGO, Campaign, Audit

Actor	Interacts With
NGO	Beneficiary, Government, Merchant
Beneficiary	Wallet, Merchant
Merchant	Wallet, Audit
Government	NGO, Policies, Analytics
Admin	Entire Platform

---

90. Business Process Summary

AidFlow AI connects all stakeholders into one continuous humanitarian execution pipeline.

Unlike traditional donation systems where transparency ends after payment, AidFlow AI maintains continuous visibility throughout:

- Campaign Creation
- Donation Collection
- Beneficiary Verification
- Aid Allocation
- Wallet Funding
- Merchant Transactions
- Proof Collection
- AI Validation
- Blockchain Verification
- Public Audit
- Impact Reporting

Every stage generates evidence, every action creates an audit trail, and every stakeholder operates under transparent, policy-driven workflows.

The result is a humanitarian infrastructure where accountability is embedded into the execution process rather than added after the fact.

SECTION F — DATABASE DESIGN & DATA ARCHITECTURE

---

91. Database Overview

AidFlow AI follows a hybrid data architecture designed to balance operational efficiency, scalability, security, and auditability.

Instead of storing all information inside one database or blockchain, the platform separates operational data from immutable verification data.

This architecture consists of:

- MongoDB (Operational Database)
- Redis (Caching & Queue Storage)
- Blockchain (Immutable Proof Layer)

Each technology performs a specialized responsibility while working together as one integrated ecosystem.

Database Philosophy

AidFlow AI follows three core principles.

Single Source of Truth

Every operational entity has one authoritative record.

Example

One donation

↓

One Donation Document

↓

Referenced everywhere else

Event Driven Architecture

Instead of continuously modifying records silently, important business events generate audit entries.

This ensures

- Traceability
- Accountability
- Historical reconstruction

Immutable Trust

Operational data may evolve.

Audit evidence never changes.

---

92. Database Technologies

MongoDB

Purpose

Primary operational database.

Stores

Users

Campaigns

Donations

Beneficiaries

Wallets

Merchants

Notifications

Policies

Reports

Analytics

Redis

Purpose

Temporary in-memory storage.

Responsibilities

Queue Management

Caching

Worker Communication

Rate Limiting

Session Storage

Blockchain

Purpose

Integrity verification.

Stores

Merkle Roots

Transaction Hashes

Audit Anchors

---

93. Collection Architecture

Major collections include

Users

Campaigns

Donations

Beneficiaries

NGOs

Merchants

Wallets

Policies

Audit Logs

Trust Scores

Notifications

Reports

Analytics

AI Decisions

Fraud Cases

Emergency Events

---

94. User Collection

Purpose

Stores every authenticated platform participant.

Fields

User ID

Role

Email

Password Hash

Verification Status

Profile

Created At

Updated At

Status

Relationships

One User

↓

One Role

↓

Many Activities

↓

Many Notifications

---

95. Campaign Collection

Stores

Campaign Name

NGO

Description

Goal

Raised Amount

Status

Category

Beneficiary Count

Documents

Trust Score

Audit Reference

Relationships

One NGO

↓

Many Campaigns

↓

Many Donations

↓

Many Beneficiaries

---

96. Donation Collection

Purpose

Tracks every financial contribution.

Fields

Donation ID

Campaign

Donor

Amount

Payment

Receipt

Audit

Blockchain Reference

Status

Relationships

One Donation

↓

One Campaign

↓

One Donor

↓

Many Audit Events

---

97. Beneficiary Collection

Stores

Identity

Verification

Wallet

Aid History

Family Details

Location

Eligibility

Trust

Relationships

One Beneficiary

↓

One Wallet

↓

Many Transactions

↓

Many Campaigns

---

98. Wallet Collection

Stores

Wallet ID

Beneficiary

Balance

Policy

Restrictions

Expiry

Status

Transactions

Blockchain Reference

---

99. Merchant Collection

Stores

Merchant Details

Business Verification

Categories

Transactions

Settlements

Trust

Audit History

---

100. Policy Collection

Purpose

Stores humanitarian policies.

Includes

Eligibility Rules

Wallet Rules

Merchant Rules

Disaster Policies

Emergency Policies

Trust Rules

Version History

Policies are immutable after activation.

---

101. Audit Collection

Purpose

Stores every important system event.

Fields

Audit ID

Actor

Action

Timestamp

Hash

Verification Status

Blockchain Reference

Previous State

Current State

102. Trust Collection

Stores

Trust Score

Reason

History

Current Level

Last Updated

Confidence

Trend

103. Notification Collection

Stores

Recipient

Message

Priority

Status

Delivery

Read Time

Created Time

104. Collection Relationships

```
Users
  │
  ├── Campaigns
  │       │
  │       ├── Donations
  │       │       │
  │       │       ├── Audit
  │       │       └── Blockchain
  │       │
  │       └── Beneficiaries
  │               │
  │               ├── Wallet
  │               ├── Merchant
  │               └── Transactions
  │
  ├── Notifications
  │
  ├── Reports
  │
  └── Trust
```

---

105. Data Flow

User Action

↓

API

↓

Validation

↓

MongoDB

↓

Queue

↓

AI

↓

Audit

↓

Merkle Tree

↓

Blockchain

↓

Notification

---

106. Indexing Strategy

Indexes

User Email

Campaign Status

Donation ID

Wallet ID

Merchant ID

Audit Timestamp

Notification Status

Trust Score

Composite indexes

Campaign + Status

Donation + Donor

Wallet + Beneficiary

---

107. Scalability Strategy

MongoDB Replica Sets

Database Sharding

Read Replicas

Connection Pooling

Caching

Queue Distribution

---

108. Backup Strategy

Daily Backup

Weekly Snapshot

Monthly Archive

Audit Backup

Blockchain Reference Backup

Encrypted Storage

---

109. Data Retention Policy

Operational Data

Active

↓

Archived

↓

Cold Storage

↓

Deleted (where legally permitted)

Audit Data

Never Deleted

Blockchain Data

Permanent

SECTION G — API & INTEGRATION DESIGN

---

110. API Architecture

Overview

AidFlow AI follows a RESTful API architecture designed around modular services, stateless communication, and standardized HTTP methods. APIs serve as the communication bridge between the frontend, backend services, AI agents, blockchain services, and external integrations.

The API layer follows these principles:

- Stateless Requests
- REST Standards
- Secure Authentication
- Versioned Endpoints
- Modular Design
- Predictable Responses
- Standard Error Handling

API Design Principles

Every API follows the same lifecycle.

```
Client Request
```

```
↓
Authentication
↓
Authorization
↓
Validation
↓
Business Logic
↓
Database
↓
Audit
↓
Queue (If Required)
↓
Response
```

API Categories

Authentication APIs

Responsible for:

- Register
- Login
- Logout
- Refresh Token
- Forgot Password
- Verify Email
- Reset Password

User APIs

Manage

- Profile
- Settings
- Notifications

- Preferences
- Activity

Campaign APIs

Support

- Create Campaign
- Update Campaign
- Delete Campaign
- Publish Campaign
- View Campaign
- Search Campaign

Donation APIs

Support

- Create Donation
- Verify Donation
- Donation History
- Donation Receipt
- Refund Status

Beneficiary APIs

Support

- Registration
- Verification
- Wallet
- Aid History
- Documents

Merchant APIs

Support

- Registration
- QR Validation
- Settlement
- Transactions
- Analytics

Government APIs

Support

- Policies
- Disaster Declaration
- Monitoring
- Reports
- NGO Verification

Administrator APIs

Support

- User Management
- Queue Monitoring
- Worker Monitoring
- AI Monitoring
- Configuration
- Audit

API Security

Every API request passes through

JWT Authentication

↓

Role Validation

↓

Permission Validation

↓

Rate Limiting

↓

Input Validation

↓

Business Rules

↓

Audit Logging

↓

Response

Response Format

Every API follows a standard response format.

Success

```
{
    "success": true,
    "message": "...",
    "data": { }
}
```

Failure

```
{
    "success": false,
    "error": "...",
    "code": "..."
}
```

Error Handling

Standard HTTP Status Codes

200 OK

201 Created

400 Bad Request

401 Unauthorized

403 Forbidden

404 Not Found

409 Conflict

422 Validation Error

500 Internal Server Error

Versioning Strategy

Future APIs use

/api/v1

↓

/api/v2

↓

/api/v3

Old versions remain available until officially deprecated.

AI Integration

Backend communicates with AI services using REST APIs.

```
Backend

↓

FastAPI

↓

Eligibility Agent

Fraud Agent

Risk Agent

↓

Prediction

↓

Response

↓

Database Update
```

Blockchain Integration

The backend communicates with blockchain services through a dedicated blockchain service layer.

```
Donation Event

↓

Audit Hash

↓
```

```
Merkle Root

↓

Smart Contract

↓

Ethereum

↓

Transaction Hash

↓

Database
```

External Integrations

Current

- Payment Gateway
- Email Service
- Redis
- Blockchain

Future

- Aadhaar APIs
- DigiLocker
- Government APIs
- SMS Gateway
- WhatsApp
- Weather APIs
- Disaster APIs
- GIS Services

API Rate Limiting

Rate limiting protects the platform against abuse.

Examples

Authentication

5 requests/minute

Donation

20 requests/minute

Public Verification

100 requests/minute

Administrator APIs

Role Based

Webhooks

Future versions support webhooks.

Examples

Donation Completed

Campaign Closed

Wallet Funded

Blockchain Anchored

Fraud Alert

Emergency Activated

API Documentation

Future releases include

Swagger

OpenAPI

Postman Collection

SDK

Developer Portal

API Summary

The REST API layer provides a secure, scalable, and modular communication interface connecting every component of AidFlow AI while maintaining consistent validation, auditing, and security across all services.

SECTION H — SECURITY, PERFORMANCE, SCALABILITY & FUTURE ROADMAP

---

111. Security Architecture

Overview

Security is integrated into every layer of AidFlow AI rather than treated as an independent component.

Security protects

- Identities
- Donations
- Beneficiaries
- Financial Transactions
- AI Decisions
- Audit Records

Authentication Security

JWT

Password Hashing

Secure Cookies

Session Timeout

Refresh Tokens

Authorization

RBAC

Permission Validation

Route Guards

Role Isolation

Data Protection

HTTPS

TLS Encryption

Password Hashing

Secure Storage

Environment Variables

Secrets Management

Input Security

Validation

Sanitization

Schema Validation

File Validation

Rate Limiting

Audit Security

Every important action creates

Audit Record

↓

Hash

↓

Merkle Tree

↓

Blockchain Anchor

Tampering becomes immediately detectable.

Threat Model

Potential threats include

Unauthorized Access

Credential Theft

Data Tampering

Replay Attacks

Fake NGOs

Duplicate Beneficiaries

Wallet Abuse

Merchant Fraud

Insider Threats

Security Response

Threat Detected

↓

Alert

↓

Audit

↓

Investigation

↓

Mitigation

↓

Recovery

---

112. Performance Strategy

AidFlow AI is optimized for high-volume humanitarian operations.

Performance considerations include

- Background Processing
- Queue Management
- Database Indexing
- Redis Caching
- Lazy Loading
- Pagination
- Compression
- Optimized Queries

Performance Targets

Authentication

< 500 ms

Campaign Search

< 1 sec

Donation Processing

< 2 sec

AI Prediction

< 5 sec

Blockchain Anchoring

Asynchronous

---

113. Scalability

The platform supports independent scaling of

Frontend

Backend

AI Services

Workers

Redis

MongoDB

Blockchain Gateway

Notification Service

Future deployments support Kubernetes and cloud-native scaling.

---

114. Monitoring & Observability

Platform health is continuously monitored.

Metrics include

CPU

Memory

API Response Time

Worker Status

Queue Length

Database Health

Blockchain Status

AI Latency

---

115. Disaster Recovery

AidFlow AI follows a layered recovery strategy.

Database Backup

↓

Configuration Backup

↓

Audit Verification

↓

Service Recovery

↓

Integrity Validation

↓

116. Current Limitations

The current implementation has several known limitations:

- AI recommendations depend on the quality and diversity of training data.
- Blockchain confirmation introduces unavoidable latency.
- Internet connectivity is required for real-time synchronization.
- Government integrations depend on external APIs and regulatory approvals.
- Wallet policies are limited to predefined rule sets.
- Advanced explainable AI and model monitoring are planned for future versions.

These limitations are acknowledged and guide future development priorities.

117. Future Scope

Future versions of AidFlow AI may include:

AI

- Disaster prediction
- Explainable AI dashboards
- Federated learning
- OCR verification
- Satellite image analysis
- Resource optimization
- Demand forecasting

Blockchain

- Stablecoin support
- DAO governance
- Layer-2 networks
- Cross-chain interoperability
- Zero-Knowledge Proofs

Platform

- Offline-first mobile applications
- Cross-border humanitarian collaboration
- Digital identity integration
- IoT-enabled supply tracking
- Volunteer management
- Real-time logistics optimization

---

118. Research Opportunities

AidFlow AI opens several avenues for further academic and industrial research:

- AI fairness in humanitarian decision-making
- Privacy-preserving beneficiary verification
- Decentralized humanitarian governance
- Transparent digital public infrastructure
- Blockchain audit optimization
- Disaster response simulations
- Human-AI collaborative governance models

---

119. Conclusion

AidFlow AI represents a shift from traditional donation platforms toward a programmable humanitarian execution infrastructure.

Rather than focusing solely on fundraising, the platform addresses the complete post-donation lifecycle through policy-driven governance, AI-assisted verification, programmable financial controls, immutable audit trails, and blockchain-backed transparency.

By combining Artificial Intelligence, modern software architecture, distributed systems, and cryptographic verification, AidFlow AI transforms humanitarian trust from subjective reputation into measurable, verifiable evidence.

The architecture is modular, scalable, and extensible, enabling future integration with emerging technologies while maintaining transparency, accountability, and security.

AidFlow AI demonstrates how intelligent digital infrastructure can improve humanitarian operations, strengthen donor confidence, empower verified organizations, and contribute to more efficient and trustworthy disaster relief ecosystems.

APPENDIX A — Glossary

Include definitions for:

- Audit Trail
- Merkle Tree
- Blockchain Anchor
- Policy Engine
- Workflow Engine
- Trust Score
- Programmable Wallet
- Queue Worker
- Eligibility Engine
- Risk Engine
- Fraud Detection
- Smart Contract
- Disaster Relief Wallet
- Public Verification
- Humanitarian Infrastructure

APPENDIX B — Acronyms

AI — Artificial Intelligence

API — Application Programming Interface

JWT — JSON Web Token

RBAC — Role-Based Access Control

NGO — Non-Governmental Organization

OCR — Optical Character Recognition

PII — Personally Identifiable Information

KPI — Key Performance Indicator

CSR — Corporate Social Responsibility

ZKP — Zero-Knowledge Proof

APPENDIX C — References

Suggested references include:

- OWASP Application Security Verification Standard (ASVS)
- OWASP Top 10
- Ethereum Documentation
- Solidity Documentation
- MongoDB Documentation
- Redis Documentation
- BullMQ Documentation
- Hardhat Documentation
- React Documentation
- Express.js Documentation
- FastAPI Documentation
- NIST Cybersecurity Framework
- ISO/IEC 27001
- UN Sustainable Development Goals (SDGs)
- UN Office for the Coordination of Humanitarian Affairs (UNOCHA)